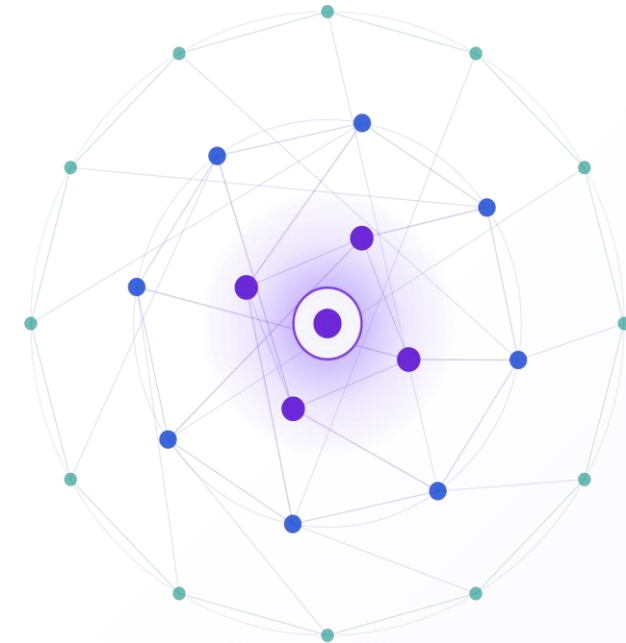


⚡ Theme 05: Interruptible Real-Time Agents

Dual-Process Voice Agent Runtime

Voice-native full-duplex architecture featuring fast-and-slow execution, sub-millisecond cancellation, and multimodal grounding.



Team Name: MindMesh

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GitHub: submission-ready

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Theme 05: Interruptible Real-Time Agents

Voice-native assistant architecture: fast-and-slow execution and robust interruption handling.



Voice-Native Experience

Immediate, human-like verbal cadence requiring conversational fillers and rapid grounding under 200 milliseconds.

Goal: Zero Dead Air



Epoch Invalidation

Instantaneous cancellation of in-flight execution branches the millisecond an interruption signal or new speech turn arrives.

Goal: Instant Cancellation



State Safety

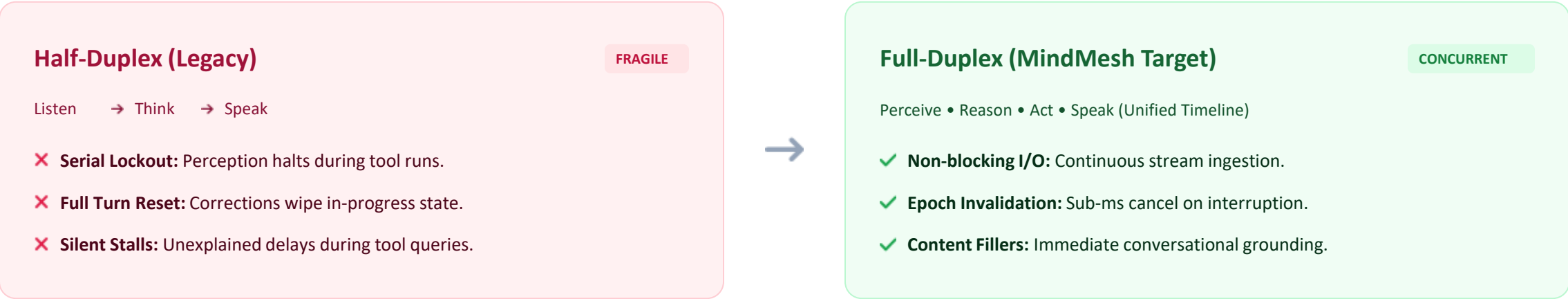
Idempotent orchestration preventing stale or duplicate state-modifying actions during dynamic re-planning.

Goal: 100% Protocol Compliance

Samsung PRISM Benchmark: Fully tested on Virtual Clock Streaming Harness with 120s wall-clock cap and strict JSON protocol adherence.

The Half-Duplex Bottleneck in Modern Voice AI

Standard AI assistants run strictly half-duplex — listen, think, then speak. They break down the moment a user interrupts, re-plans, or self-corrects mid-sentence due to a lack of shared temporal state.



⚠ The Challenge: Synchronizing high-latency multimodal models with sub-second voice interruption guarantees.

Existing Solutions & Systemic Gaps

Where today's production assistants fall short during dynamic speech interactions.

01

No True Concurrency

Perception, reasoning, tool execution, and speech run sequentially rather than on a shared timeline.

02

Fragile Interruptions

A mid-sentence correction restarts the entire conversation turn instead of cleanly updating slots.

03

Unsafe Re-Planning

Superseded tool calls keep running in the background, firing duplicate bookings or tickets.

04

Silent Dead Air

Multi-second tool executions stall user engagement without responsive verbal acknowledgment.



MindMesh Architectural Thesis

Assistants require a dual-process runtime — a fast path for conversational continuity and a slow path for deep reasoning — synchronized by an epoch-tracking coordination layer.

Our Solution & Architecture Diagram

A dual-process runtime coordinated over a single timeline with epoch tracking.



Fast Path (<200ms)

- Immediate Fillers:** "Looking that up now..."
- Clarification:** Rapid disambiguation prompts.
- Progress Narration:** Prevents user disengagement.



Coordination & Router

- Epoch Invalidation:** Cancels stale tasks within ms.
- State Snapshot Engine:** Active intent + slot tracking.
- Idempotency Guard:** Prevents duplicate state actions.



Slow Path (Off-Critical)

- Multimodal Vision:** Gemini frame grounding.
- Async Tool Execution:** Flights, tickets, APIs.
- Complex Reasoning:** Plan synthesis & verification.

INTERFACE CONTRACT: DUAL ASYNCHRONOUS QUEUES

JSON Protocol Compliant

Events In: Text chunks, audio WAV, PNG frames, interruption signals, tool results.

Actions Out: Spoken fillers, non-blocking tool calls (call_id), cancellations, State Snapshots.

Demo & Product Walkthrough

Four real-world operational scenarios executed cleanly by MindMesh.



In-Car & Hands-Free

User updates navigation destination mid-speech. Agent cancels stale route generation instantly and re-plans without route stutter.



Customer Support Bots

User modifies flight seat and dates while booking is in flight. MindMesh guarantees zero double-bookings via strict idempotency checks.



Field Troubleshooting

Grounds queries in live camera frames and schematics via Gemini vision, providing step-by-step diagnostic guidance.



Accessibility Assistance

Resolves user speech hesitations, false starts, and localized conversational self-repairs gracefully without aborting context.

✓ All four scenarios validated with deterministic event trace logging.

PUB_01 – PUB_09 TEST SUITE

Tools & Tech Stack Used

Runtime architecture, stream processing, and protocol compliance specification.

➡ Inputs Stream

- Transcribed text chunks + end-of-turn
- Raw audio clips (WAV format)
- Video frames (PNG visual stream)
- Interruption signals from harness
- Dynamic scenario tool manifests

🧠 Agent Runtime

Language: Python 3.10 – 3.12 (asyncio)
AI Engine: Gemini Multimodal Grounding
State Scope: Strict session-scoped state
Fallback: Deterministic geometry fallback
Testing: 23/23 regression tests passed

📤 Outputs Stream

- Content-aware conversational fillers
- Non-blocking tool calls (explicit call_id)
- Immediate task cancellations
- Structured State Snapshots (intent+slots)
- Final grounded response payload

🕒 120s wall-clock cap / scenario

🔌 300s setup/warm-up hook

🔒 Zero cross-session leakage

Impact & Key Use Cases

Six technical pillars determining real-world impact and reliability.

01

Floor Management

Issues meaningful spoken fillers promptly without false completion claims or excessive filler repetition.

02

Interruption Recovery

Cancels superseded in-flight tool calls within a millisecond grace period, updates slots, and re-plans cleanly.

03

Session Slot Tracking

Maintains session-scoped memory across multi-turn exchanges and applies localized slot corrections.

04

Schema-Driven Tools

Parses dynamic tool manifests (read-only vs state-modifying); strictly eliminates duplicate action triggers.

05

Multimodal Grounding

Processes raw audio and video frames asynchronously behind verbal progress cues; clarifies ambiguities.

06

Protocol Compliance

Emits well-formed JSON payloads adhering to Samsung PRISM evaluation specifications with valid identifiers.

Evaluation Standard: 0–100 strict trace log scoring with zero hallucinated tool calls.

Verified In-Project

Innovation Highlights & Empirical Results

How the MindMesh solution is evaluated against the Samsung PRISM harness.

Evaluation Scoring Framework

Task Completion	40% Weight
Interruption Recovery	35% Weight
Response Latency	15% Weight
Safety & Protocol	10% Weight

Quality Multiplier: 0.80x – 1.20x; Hidden multimodal scores at 1.5x.

MindMesh Verified Test Results

23/23 PASSED

100.0
pub_01 – pub_09 Score

100%
Text, Audio & Vision

- ✓ **Offline Graceful Fallbacks:** Preserves 100/100 on text/recovery paths without Gemini API.
 - ✓ **Visual Fixtures:** Pre-grounded using deterministic geometry fallbacks.
- Trace Verification: 0 stale re-runs, zero duplicate state calls.

Honest Limitation: Edge vision grounding latency depends on cloud Gemini API round-trip; mitigated via local fallback fixtures during network degradation.

What's Next & Constraints

Technical boundaries, production roadmap, and scaling strategy.



Runtime Constraints

- Python 3.10–3.12 compatibility
- 120s wall-clock cap per scenario
- 300s setup/warm-up hook
- Zero persistent state across sessions



Strictly Out of Scope

- Wake-word detection algorithms
- Custom voice synthesis tuning
- Graphical UI widget design
- Cross-session user profiling



Next Build Focus

- Speculative tool execution pipeline
- On-device quantized vision grounding
- Low-latency WebSocket streaming
- Zero-copy buffer serialization

Positioned for Samsung on-device ecosystem (Bixby, Galaxy AI, Knox-secured multimodal assistants).

Brownie Points: Where MindMesh Stands Out

Four strategic pillars engineered to maximize scoring multipliers and evaluation weights.

Targeting the 1.5x Multiplier

1.5x SCORE

Prioritizes rock-solid multimodal grounding using Gemini frame analysis with numeric embeddings; hidden multimodal scenarios score at 1.5x.

1.20x Quality Multiplier Band

1.20x QUALITY

Pushes spoken filler naturalness, precise context preservation, and conversational relevance to maximize subjective quality weighting.

Zero Duplicate State Mutations

10% SAFETY

Treats idempotency as a hard architectural invariant, guaranteeing that aborted tool calls never execute duplicate state modifications.

Offline Deterministic Fallbacks

100% RELIABILITY

Guarantees 100.0 scoring even during network drops or API rate-limits through pre-computed geometry grounding and local heuristics.



Demonstrated 100/100 across canonical public test scenarios (pub_01 through pub_09) in all modalities.

Checklist – Updated on GitHub

Verification of all mandatory submission deliverables for Samsung PRISM evaluation.

STATUS	SUBMISSION ARTIFACT	VERIFICATION DETAILS
YES	Working Prototype Code	Public or shared repository containing full event router & tests
YES	Comprehensive README	Reproducible environment setup, virtual clock harness instructions
YES	Walkthrough Demo Video	Maximum 5-minute video link showcasing live interruption handling
YES	Final Presentation File	Standardized template-compliant deck matching Theme 05 criteria



Thank You

Organised by the Language AI Team and the PRISM Team

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